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Automatic Railway Gate Control System Using PLC

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Abstract: The aim of this project is to develop a prototype that control the railway gate using PLC (PROGRAMMABLE LOGIC CONTROLLER). Whenever train touches base at the IR sensor, caution is activated at the railway crossing so that the general population get instruction that entryway will be shut. At that point the control module initiates and shuts the gates on either side of the track. Once the train crosses, this module naturally lifts the gate. For mechanical operation of a gate DC adapted engines are utilized. We are using PLC as a main control unit. As per the instructions produced at the PLC, the proper action (i.e., shut or lift) will be made. This logic was implemented in PLC (Programmable Logic Controller)

Keywords: PLC, IR sensor, DC Motor

I. INTRODUCTION

When it involves public and private safety, it is always desirable to improve the signaling system and the way people are warned about potentially dangerous situations. The existing conventional signaling system most of the times rely on the oral communication through telephonic and telegraphic conversation as input for the decision making in track allocation for trains. There is a large scope for miscommunication gap due to the higher human interference in the system. At railway crossing, the traditional railway system uses the warning lights and human controlled gates to alert the people about an oncoming train. When the train leaves the station, the station agent informs the gatekeeper regarding the arrival of the train using the telephone and telegraph. The gatekeeper closes the gate depending on the timing at which the train arrives. Hence, if the train is late because of certain reasons, then gate stay closed for an extended time inflicting traffic near the gates. The automatic railway gate control at the railway crossing the arrival of the train is detected by the sensing element placed on the track at a precise distance from the gate. This sensing element detects the approaching train and consequently controls the operation of the gate. When the wheels of the train moving over the track there will be position switch and it will send the signal to PLC to indicate train arrival. This reduces the time that gate is closed as compared to the gates operated manually. In addition, this also reduces the human labor. This sort of automatic railway gate system is employed in associate unmanned gate crossing where the probabilities of accidents are higher and reliable operation is needed.

II. LITERATURE REVIEW

The automatic railway gates operation has been projected using various methods. As proposed by Xishi Wang (1992), the process of developing fault tolerance method has been applied for both the hardware and the software components. Magnetic sensors placed underground to detect the train are less affected by environmental changes and recognizes the direction of movement of vehicles.

Jeong Y (2008) defined the railway auto control system using OSGi and JESS. The state of railway cross has been estimated using JESS in the technique. The issues in the technique are the insufficient inline citations and also multiple issues related to OSGi. The different methods used by locomotive pilots which can avoid the accidents and the safety measures while crossing the level crossings are also discussed. Atul Kumar Dewangan (2012) gave a detailed introduction about the present railway technology and also discussed the disadvantages of manually activated railway signals and the railway warnings at the level cross. The train detectors act as the major component in the train automation system.

Banuchander J (2012) developed a method to concentrate on anti-collision system to identify the collision points and to report these error cases to main control room, nearby station as well as grid control stations. Efficient Zig-Bee based Train Anti-Collision using Zig-Bee technology for railways is implemented.

Greene R.J. (2006) anticipated an intelligent railway crossing control system for multiple tracks that features a controller which receives messages from incoming and outgoing trains by sensors. These messages contain detail information including the direction and identity of a train. Depending on those messages the controller device decides whenever the railroad crossing gate will close or open. But this technique has the issue of high maintenance.

Kawshik Shikder (2014) projected the automatic operation of railway gates using RF technology. The major issue of this technique was every train could be provided with RF technology. Thus it was economically feasible to implement. Anil M.D. (2014) proposed advanced railway accident prevention system using sensor networks. He used ZigBee RF module to communicate between base station and trains. But, ZigBee was a short distance communicating device. Therefore it is practically not possible to implement this technique motor looks as a straight expansion over the simple DC ones.

III. DESCRIPTION AND WORKING

The main idea of our project is to develop the automatic to control the railway gate and traffic light at level crossing using PLC. As a train approach the railway crossing from either side, the sensors placed in the track at a certain distance from the gate detects the approaching train and accordingly controls the operation of the gate. The proposed system uses infrared sensors to detect the trains crossing the road and motor to control traffic light and the opening or closing of gates. The system uses two sensors to detect the arrival of the train and a second sensor to detect the leaves of the train. When the arrival of the train is sensed, signals are provided to the traffic indicating the arrival of the train on the road such as the signal turns red and the motor operates to close the gate. When the second sensor detects the train, then the signal turns green and the motor operates to open the gate. The gate remains closed until the train completely moves away from the crossing road. Thus, automation of the gate operations at the level crossing is achieved using sensors, motor and traffic light. The signal from the colour resistive sensor will play an important role in gate control and traffic light control process. When the PLC receives the signal from the sensor, then it produces the output based on ladder program which is fed to the stepper motor driver for closing the gate. When the traffic light indicates the red signal for the vehicles that passing through the road. Similarly, when the PLC receives the signal from the receiver it indicates the train indicates the train has passed away SWITCH gate is in open position which indicates that by a green signal for the level crossing.

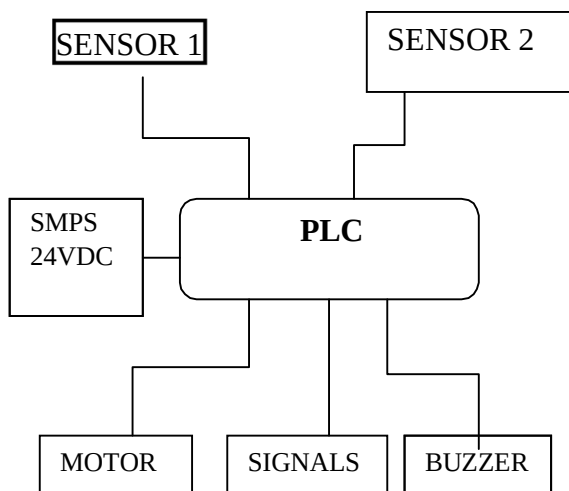
A. PLC

PLC defines as a “digitally operating electronic apparatus which uses a programmable memory for the internal storage of the instructions for implementing specific functions such as logic, timing, counting and arithmetic, sequencing to control through digital or analog input/output modules, various types of machines or processors.”

B. DC Motor

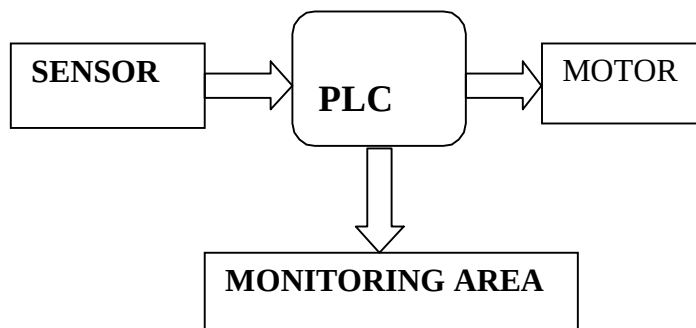
Geared DC Motors can be defined as an extension of DC motor which already had its insight details demystified here. A geared DC Motor-30RPM, 12Volts geared are generally a simple DC Motor with a gearbox attached to it. The speed of the motor is counted in terms of rotations of the shaft per minute and is termed as RPM. The gear assembly helps in increasing the torque and reducing the speed. Using the correct combination of gears in a gear motor, its speed can be reduced to any desirable figure. This concept where gears reduce the speed of the vehicle but increase its torque is known as gear reduction. This insight will explore all the minor and major details that make the gear and hence the working of geared DC motor. At the first sight, the external structure of a DC geared

IV. BLOCK DIAGRAM



The above architecture shows the arrangement of the components and connected in such a manner. Here each and every component has connected in such a way to communicate and to do control functions of all the components. By having a monitoring area all the process can be monitored at a single place. PLC is placed in central because it does all the functions according to sensor inputs.

V. SYSTEM ARCHITECTURE



Here we can see the complete flow process of the project. The signal from the sensor will play a major role in the complete process as an initial state. The signal is sent to the PLC, then it produces the output based on our ladder program. It is fed to drive the DC motor, which only needs the pulses. As a monitoring area, we will have some display arrangements to monitor the complete process. Here, the signal to the monitoring area will be taken from the PLC.

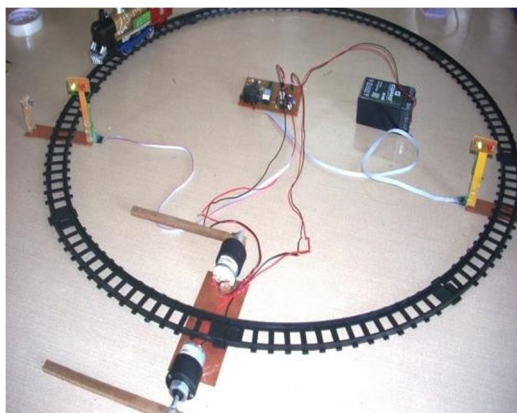
VI. ADVANTAGES

- A. Saving of money
- B. Move towards smart city
- C. Accident avoidance
- D. Human safety

VII. CONCLUSION

Nowadays, PLCs are being used to reduce maintenance and labor cost of many control systems solutions. Within the next five to ten years, many existing control systems will be replaced by PLCs. Therefore, it is highly recommended that a complex system like railway gate controlling will be based on PLCs. In this project, we have developed such a control system. A prototype road and rail line model with railway gate has been created and implemented. After successful testing of the whole system, it was found that the developed system operates satisfactorily.

VIII. SYSTEM MODEL



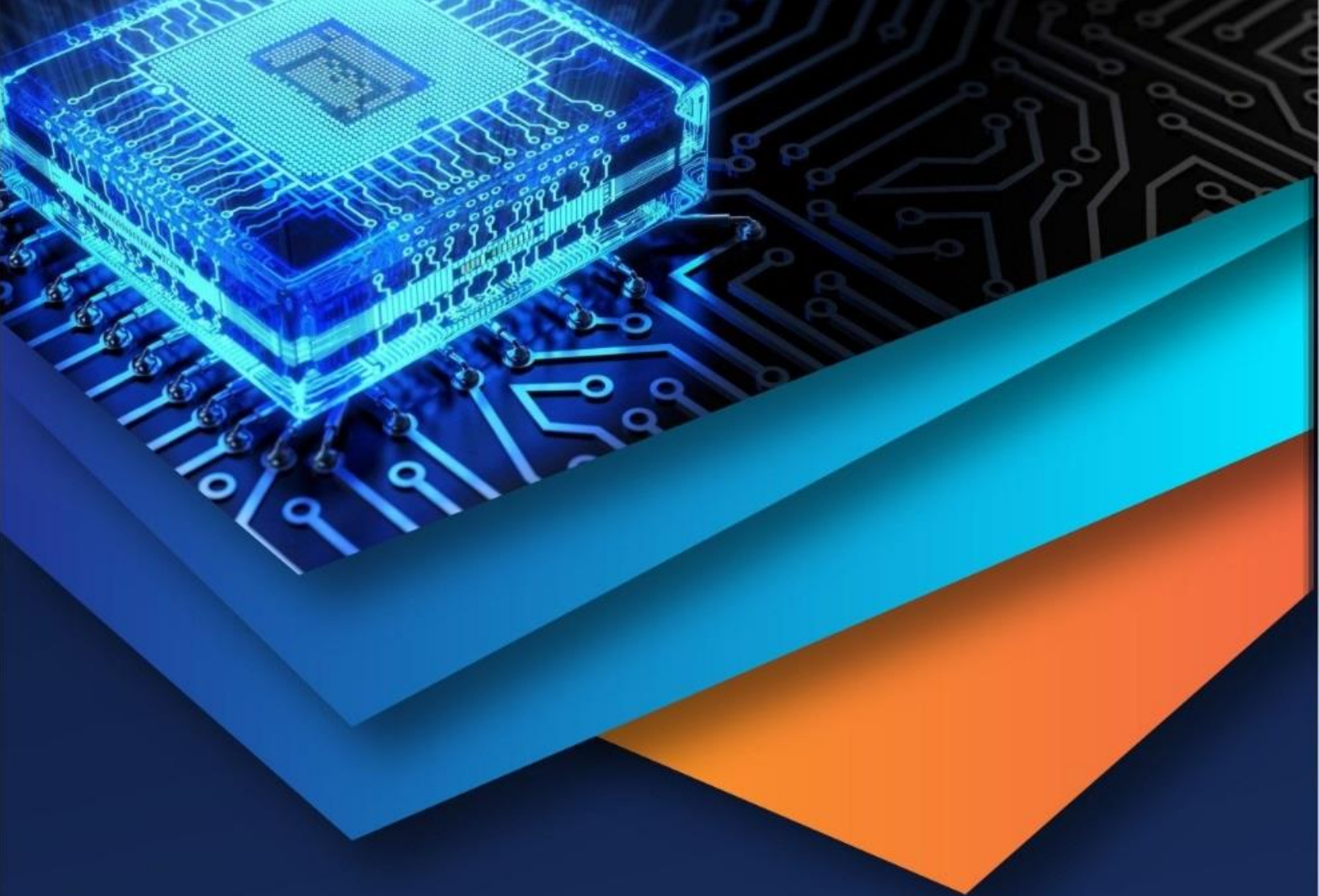


VIII. ACKNOWLEDGEMENT

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